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Technical Q&A
for NS1 NEON SMBA PRD
NS1-SMBA-REQ-0001 Rev E

Below are questions and answers related to the Sounder for Microwave-Based Applications (SMBA) Performance Requirements Document (PRD), Revision E. In the event of any discrepancy, the final Request for Proposal provisions and contract terms will take precedence.

Question: How is the $NEDT$ calculated for heritage and non-heritage channels? How is the $NEDT_{1/f}$ calculated?

Answer: Microwave radiometer total noise $NEDT$ is commonly dominated by thermal noise $NEDT_{thermal}$ and 1/f noise $NEDT_{1/f}$ and may be calculated by:

SMBA PRD Equation 5: $$NEDT = \left[(NEDT_{thermal})^2 + (NEDT_{1/f})^2\right]^{1/2}$$

The thermal noise, $NEDT_{thermal}$, can be calculated by:

$$NEDT_{thermal} = \frac{Thermal\ Noise\ Contribution\ to\ NEDT}{\sqrt{B \times \tau}}$$

Where the "Thermal Noise Contribution to $NEDT$" is the value in the 4th column of Table 4.1-4, B is the channel bandwidth (the sum of all passbands in the channel) in MHz, and τ is the integration time in ms.

$NEDT_{thermal}$ is then scaled for the SMBA design and sets the $NEDT_{thermal}$ limit.

For heritage channels, the channel bandwidth and the $NEDT$ requirements are listed in Table 4.1-3 and the integration time is established by the SMBA proposer's design.

For non-heritage channels, integration time and channel bandwidth is established by the SMBA proposer's design and the resulting $NEDT$ is reported in CDRL documentation.

For $NEDT_{1/f}$, SMBA-30 applies to both heritage and non-heritage channels.

The 1/f noise ratio, $R_{1/f}$, can be calculated from

SMBA PRD Equation 6: $R_{1/f} = \frac{(NEDT_{1/f})^2}{NEDT^2}$

Example 1: Thermal Noise Contribution to $NEDT = 21.6K$ taken from Table 4.1-4 for V-band < 56 GHz, $B = 180\ MHz$, and using $\tau = 18\ ms$ as an example, from the $NEDT_{thermal}$ equation above, the resulting $NEDT_{thermal} = 0.3795K$.

Example 2: From Example 1, using $R_{1/f}$ = 0.1 as the max allowed by SMBA-30, with $NEDT$ = 0.4 K from Table 4.1-3, SMBA PRD Equation 6 can be used to solve for $NEDT_{1/f}$.

$$NEDT_{1/f} = \sqrt{R_{1/f}} \times NEDT = 0.1265K$$

Example 3: Using the results from Examples 1 & 2, the $NEDT$ can now be computed using SMBA PRD Equation 5.

$$NEDT = \left[(NEDT_{thermal})^2 + (NEDT_{1/f})^2\right]^{1/2} = [(0.3795)^2 + (0.1265)^2]^{1/2} = 0.4K$$

Question: Does the spatial horizontal resolution requirement (SMBA-44) apply to instantaneous footprint or aggregated footprint?

Answer: The spatial horizontal resolution requirement (SMBA-44) applies at an instant of time (i.e., Instantaneous Field of View) rather than over the time duration of a footprint.

IFOV is the portion of the Earth's surface enclosed by the projected -3dB antenna contour at an instant of time (i.e., instantaneous footprint). An IFOV does not include smearing due to scanning or satellite orbital movement, nor any aggregating or spatial resampling of different footprints.

SMBA follows traditional practice of using the IFOV to check compliance with requirements for imaging without spatial gaps in the along-scan or along-track directions. The SMBA footprint size requirements (SMBA-42 and SMBA-44) must be met at the IFOV level at nadir.

Question: Is compliance to the new passband gain ripple requirement (SMBA-41) constrained to front end hardware design?

Answer: SMBA-41 is agnostic of hardware design and/or software design choices employed to comply.

Question: Will the SMBA instrument be powered ON or OFF during launch?

Answer: PRD is written to be inclusive of instrument designs that require being powered ON and instrument designs that require being OFF at launch.